Footstep Power Generation and Activity Detection using Piezo Sensors with Machine Learning – Report

GROUP-10

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ABSTRACT

This project focuses on generating electrical energy using piezoelectric sensors embedded beneath a surface and integrating machine learning for human activity detection. The system captures mechanical stress from footsteps, converts it into electrical signals, and processes the data using an Arduino Uno and Python-based ML model.

INTRODUCTION

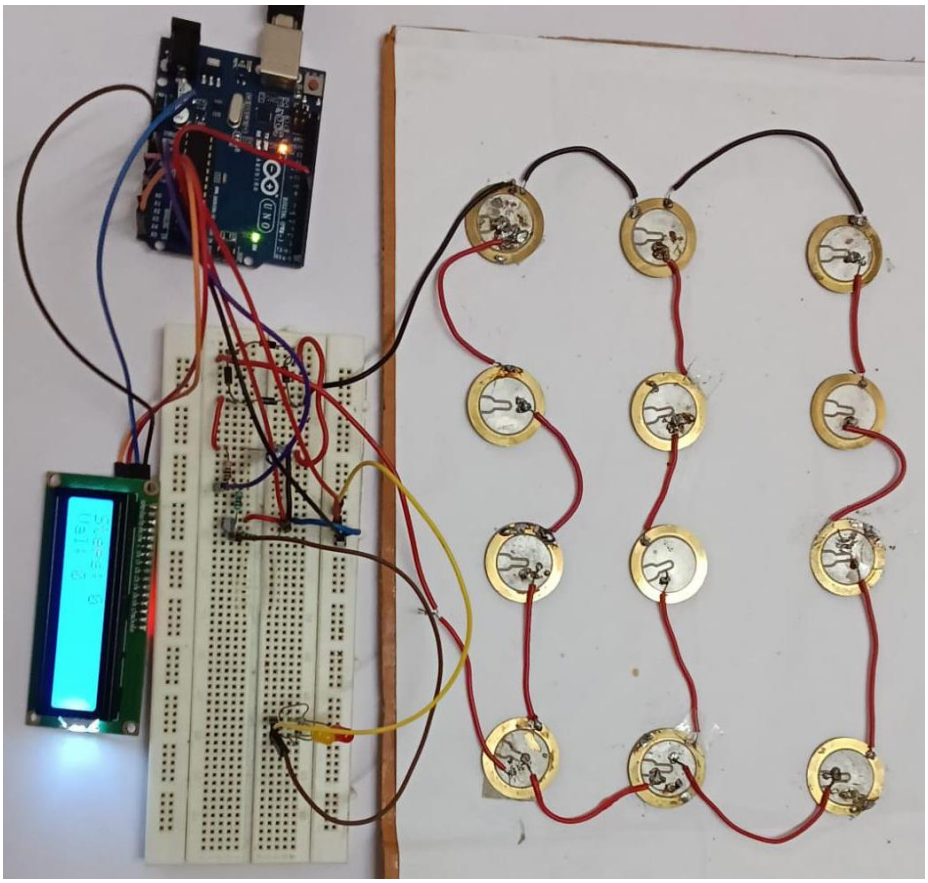
Energy harvesting from human movement is an emerging field in sustainable technology. Piezoelectric sensors provide a simple and effective way to convert mechanical vibrations into electrical energy. This project combines embedded systems and machine learning to not only generate energy but also classify human activities such as walking, running, jumping, and standing.

OBJECTIVES

1. To generate electrical energy using piezo sensors.
2. To design a sensor grid using series-parallel connections.
3. To collect real-time data using Arduino.
4. To apply machine learning for activity classification.
5. To visualize sensor behavior using real-time graphs.

SYSTEM ARCHITURE

The system consists of:
- Piezo sensor array
- Arduino Uno
- Signal conditioning circuit
- Serial communication interface
- Python ML model
- Visualization system



WORKING PRINCIPLE

When pressure is applied to piezoelectric sensors, they generate voltage due to deformation. The sensors are arranged in a series-parallel network to increase voltage and current output. The Arduino reads analog values and sends them to a computer for processing. A machine learning model extracts features and classifies activities.

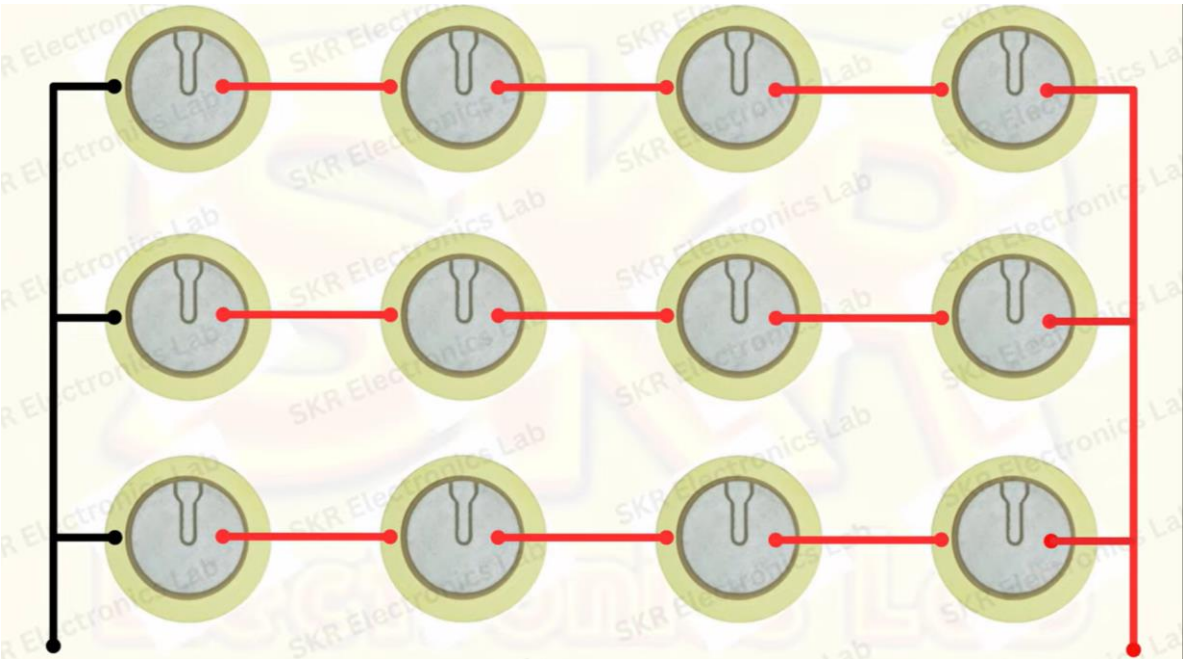
- **Piezoelectric Effect:** When pressure is applied to the piezoelectric sensors, they deform, generating an AC voltage proportional to the applied force.

- **Rectification:** The AC voltage is then passed through a bridge rectifier, converting it into pulsating DC.
- **Filtering:** A filter capacitor smooths out the rectified DC, providing a more stable voltage for subsequent stages.
- **Energy Storage:** The conditioned DC energy is stored in a battery or supercapacitor, ensuring a continuous power supply even when no footsteps are present.
- **Arduino Integration:** The Arduino microcontroller continuously reads the voltage levels from the storage unit and processes signals from the sensors to count footsteps.
- **Output Display:** The measured voltage and accumulated footstep count are displayed on an LCD or indicated using LEDs, providing real-time system feedback.

SENSOR ARRANGEMENT

Vertical connections (top to bottom) are in series to increase voltage output. Horizontal connections (left to right) are in parallel to increase current and coverage area.

In my project, piezo sensors are connected in a hybrid configuration where each vertical column of sensors is connected in series to increase voltage, and multiple such columns are connected in parallel to increase current and improve area coverage.



DATA COLLECTION AND PROCESSING

Sensor data is collected in real-time via serial communication. Data is segmented into windows and processed using feature extraction techniques such as amplitude, standard deviation, energy, peak count, and zero crossings.

MACHINE LEARNING MODEL

A **Random Forest classifier** is used for activity detection. The model is trained using labeled datasets for walking, running, jumping, and standing. The extracted features enable the model to differentiate between activities.

Activity	Amplitude	Energy	Peaks
Standing	~0	~0	0
Walking	Low	Low	Few
Running	Medium	Medium	More
Jumping	High	Very High	Sharp

DATASET PREPARATION

The dataset for this project was created by collecting real-time sensor data from the piezoelectric sensors using an Arduino Uno. The Arduino continuously reads analog values from the piezo sensors and transmits them through serial communication.

For each activity—walking, running, jumping, and standing—the corresponding sensor values were recorded using the serial monitor. These readings were then stored in separate text files named ***walking.txt*, *running.txt*, *jumping.txt*, and *standing.txt*.** Each file represents labeled data for a specific activity, which is later used for training the machine learning model.

DATA PREPROCESSING

The collected raw data was segmented into smaller windows of fixed size (20 samples per segment) to capture short time-based patterns in the signal. Each segment represents a small portion of activity data.

From each segment, important statistical features were extracted, including amplitude, standard deviation, energy, peak count, and zero-crossing rate. These features help in summarizing the signal characteristics and make it easier for the machine learning model to distinguish between different activities.

Model Training

A **Random Forest classifier** was used for training the model. The extracted features from all activity datasets were combined into a structured dataset, where each sample contains feature values along with its corresponding activity label.

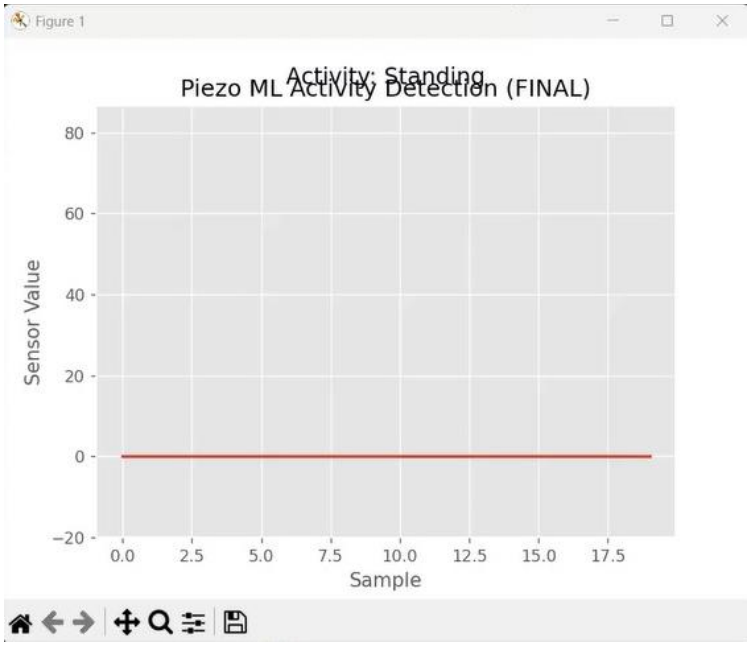
The dataset was divided into input features and target labels. The model was then trained using these features, enabling it to learn patterns associated with different activities such as walking, running, jumping, and standing.

The trained model is capable of predicting the activity based on new incoming sensor data in real time.

RESULT AND OBSERVATIONS

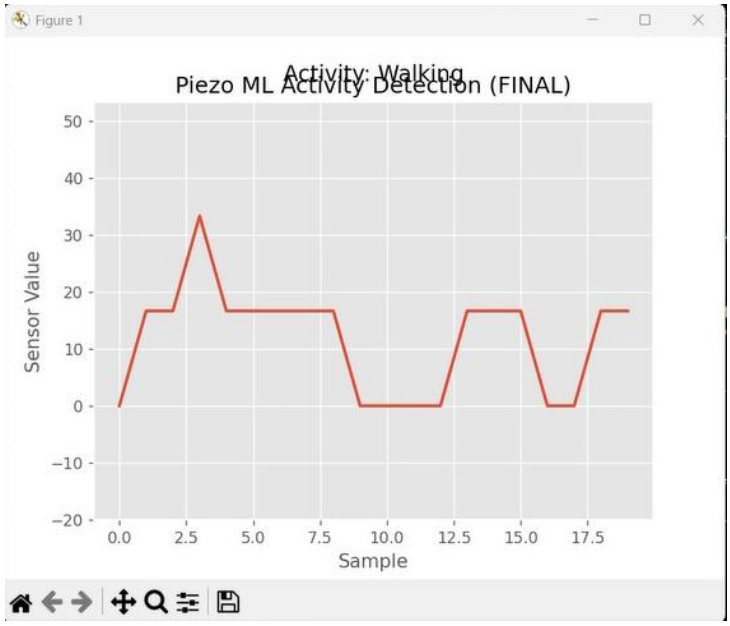
- Standing produces near-zero output.
- Walking generates small periodic spikes.
- Running produces higher frequency signals.
- Jumping results in high amplitude spikes.

STANDING



The graph shows near-zero sensor values with minimal variation, indicating no significant pressure or movement on the piezo sensors. Since piezoelectric sensors respond only to changes in force, a stationary condition produces almost no output. The small spike observed is due to minor disturbances or noise. Hence, this pattern corresponds to the standing activity.

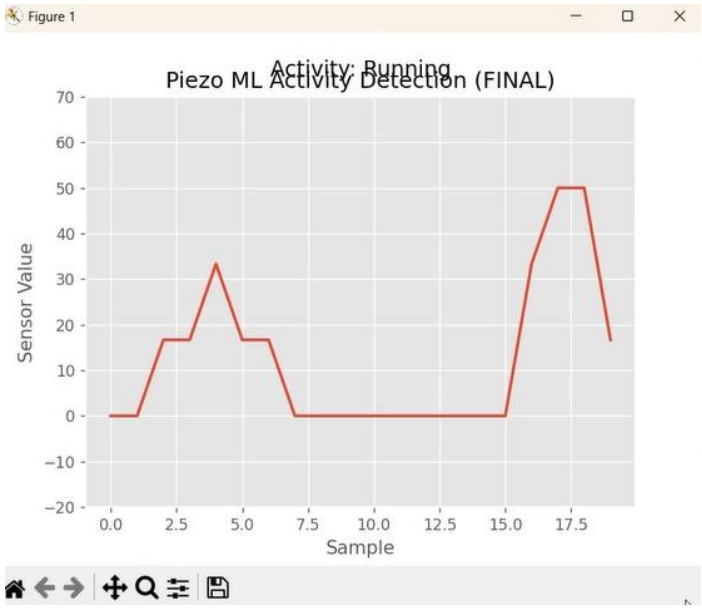
WALKING



The graph shows moderate, periodic spikes in sensor values, which correspond to footsteps during walking. Each spike represents a footstep impact, while the intervals of low or zero values indicate the time between steps. Compared to other activities, the amplitude is neither too high nor too low, and the frequency of spikes is consistent. This pattern is characteristic of walking, where pressure is applied rhythmically but not very forcefully.

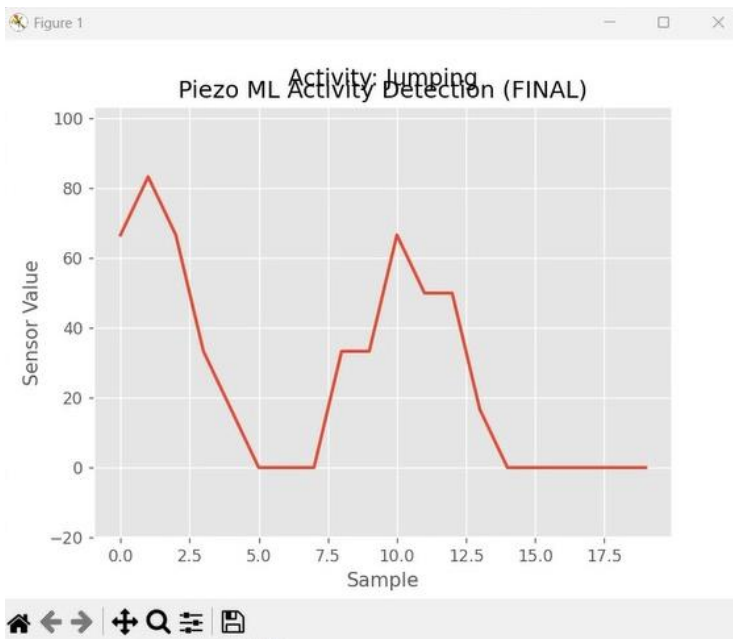
periodic (step → gap → step)

RUNNING



The graph shows higher amplitude spikes and relatively faster variations compared to walking. These spikes represent stronger and quicker foot impacts on the piezo sensors. Running involves greater force and shorter intervals between steps, which results in increased signal energy and more prominent peaks. The presence of higher peaks and quicker transitions between values clearly indicates a running activity.

JUMPING



The graph shows very high amplitude spikes with sudden rises and drops, indicating strong impact forces on the piezo sensors. Jumping involves powerful actions such as take-off and landing, which generate large amounts of energy compared to walking or running. The sharp peaks and higher signal values clearly represent these intense impacts, while the intervals of low values indicate the time when the feet are off the ground. This pattern is characteristic of jumping activity.

RESULT AND OBSERVATIONS

- Standing produces near-zero output.
- Walking generates small periodic spikes.
- Running produces higher frequency signals.
- Jumping results in high amplitude spikes.

CHALLENGES

- Noise in sensor readings
- Signal instability
- Variability in human motion
- Hardware sensitivity

LIMITATIONS AND OBSERVATIONS

During the experimentation, it was observed that a relatively high amount of force was required to generate noticeable output from the piezoelectric sensors. This is primarily because piezo sensors respond more effectively to sudden mechanical deformation or vibrations rather than slow or static pressure.

In the current setup, the sensors are placed directly on a rigid surface without any mechanical amplification mechanism. As a result, the applied force is not efficiently transferred to the sensors, leading to lower sensitivity. Additionally, the series connection of multiple piezo elements increases the voltage but reduces the responsiveness to small pressure variations.

To address this limitation, mechanical components such as springs or elastic layers can be introduced in the structure. The use of springs helps in converting applied pressure into dynamic deformation, allowing the piezo sensors to bend more effectively and generate higher output even for smaller forces.

REAL-LIFE APPLICATIONS

- Smart floors in malls and airports
- Energy harvesting in railway stations
- Fitness tracking systems
- Security monitoring systems
- Rehabilitation and healthcare monitoring

GLOBAL USAGE

Countries like Japan, UK, and USA have implemented piezoelectric flooring in high-footfall areas such as train stations, airports, and public walkways to generate renewable energy and monitor crowd movement.

SURFACE IMPLEMENTATION

To install this setup in real life:
- Use a flexible but strong top layer (rubber or acrylic)

- Place piezo sensors beneath pressure points
- Use foam or springs to distribute force evenly
- Ensure proper insulation and wiring safety

CONCLUSION

This project successfully demonstrates the integration of hardware and machine learning for energy harvesting and activity detection. It highlights the potential of piezoelectric systems in smart environments and sustainable energy solutions.